

Rahnuma

रहनुमा رہنما

A working guide to the ILM, PANINI and Zistgah stack, from bytes and scripts to compilers, Pāṇini and quantum substrates

प्रकृति

	written	Devanagari hub	Romenagri	and back
Devanagari	प्रकृति	→ प्रकृति	→ prak_ri_ti	→ प्रकृति
Telugu	ప్రకృతి	→ प्रकृति	→ prak_ri_ti	→ ప్రకృతి
Bengali	প্রকৃতি	→ प्रकृति	→ prak_ri_ti	→ the same identity
Urdu, as usually written	ہندوی	→ हनदवी	→ _hana_dawee	→ हनदवी
Urdu, with zer and sukun	ہندوی	→ हिन्दवी	→ _hin_dawee	→ हिन्दवी

One identity, many scripts. Telugu and Bengali meet Devanagari in one hub, become the same Romenagri word, and come back exactly. Urdu meets the hub at what is written: without its short-vowel marks the word reads hanadawee; with zer and sukun it is hindawi. Every cell here was produced by the Hindawi tools while this guide was built.

Abhishek Choudhary, AyeAI. Version 2.0.0, 3 October 2026.

Rahnuma, the edition for the field: Education

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Part I: Orientation

1. How to use this guide

In plain words. This guide explains a large body of work on computers, languages and intelligence. You can read it, run the programs on your own computer, or ask an AI to help you, and every example in it was really run.

This guide is for anyone who wants to use, check or extend the work gathered in the *zistgah*, *project-ilm* and *hindawiai* organisations on GitHub: students preparing for GATE, engineers who write compilers, firmware or chip layouts, linguists who work with scripts and sound, and researchers in quantum computing and AI. It asks for curiosity and a laptop, and nothing else.

There are four ways in.

- **Understand.** Part II gives the whole architecture first: where the work comes from, how the parts compose, each term with its nearest analogue and the difference, and the three Acts.
- **Read.** Parts IV and V explain the ideas the work rests on, with the mathematics written out: how numbers and letters sit in memory, what makes a transliteration reversible, how Pāṇini's grammar computes, how a compiler turns text into machine instructions, and how an oscillator network and a quantum circuit attack the same problem.
- **Run.** Part III gives the exact commands to set up Linux, Windows with WSL, or macOS, and to run every component of the estate on your own machine.
- **Ask.** Chapter 24 shows how to put any AI assistant to work on these repositories, and how to check what it tells you.

Conventions used throughout:

- Every example that could run while this guide was being built did run, and the text shows its real output, marked *Executed while this guide was built*. The package's own `gate`, `bash ops/verify.sh`, runs those examples again and fails if any output differs from what is printed here.
- Status words are used exactly. **Tested:** a test in the repository asserts it. **Executed:** it ran and its output is shown. **Written:** the code exists but has not been run here. **Mocked:** a stand-in replaces hardware that is not attached. **Planned:** described, not built.
- Everything said about the repositories was retrieved from the repositories themselves on 29 September 2026: their listings on GitHub, their `CITATION.cff` files and their READMEs. Where a repository does not record something, this guide says so instead of filling the gap.
- A dagger, †, marks a statement that follows the author's working notes and write-ups as supplied on 29 September and 3 October 2026, rather than the repositories.
- Sanskrit is transliterated in IAST. The spelling is British.

The work lives at github.com/zistgah, github.com/project-ilm and github.com/hindawiai, on the site zistgah.org, and in the Zenodo records listed in Chapter 12.

3. Start here: find your way in

In plain words. Four steps to find your part of the guide: who you are, your board or field, what you want to do, and your path with a PDF made for you.

This guide serves a school student and a senior researcher; a village health worker and an investor. Nobody needs all of it. Four steps find your part, on the site or with this page:

1. **Who you are.** A school student, a college student, a postgraduate or doctoral researcher; a teacher; an academic researcher; a working professional, someone in public service, policy or law, someone in community, NGO or village work, an artist, writer or maker; an industry partner or investor; or an AI agent working for a person.
2. **Your board or your field.** A school student chooses a board, ICSE, ISC, CBSE, IGCSE, IB, a state board or another; and a class. Everyone else chooses a field from the international standard classification of fields of study, ISCED-F 2013, the same one FAKIR's lattice uses. It places each subject where its own scholars would: political science is under social and behavioural sciences, not the humanities; law is under business, administration and law; linguistics is under languages.
3. **What you want to do.** Learn the basics, run the software, study or research, teach, contribute, or adopt, fund or invest.
4. **Your path and your edition.** The site lists your chapters in order, with reading times and the components to try first, and offers a PDF edition made for you. Every edition below is a printed selection from this one guide, so all of them stay in step.

Edition	PDF
School student	role-school.pdf
College student	role-college.pdf
Postgraduate or doctoral researcher	role-pg.pdf
Teacher or lecturer	role-teacher.pdf
Academic researcher	role-researcher.pdf
Working professional or engineer	role-professional.pdf
Public service, policy or law	role-public.pdf
Community, NGO or village work	role-community.pdf
Artist, writer or maker	role-creator.pdf
Industry partner, investor or funder	role-industry.pdf
Generic programmes and qualifications	field-00.pdf
Education	field-01.pdf
Arts and humanities	field-02.pdf
Social sciences, journalism and information	field-03.pdf

Edition	PDF
Business, administration and law	field-04.pdf
Natural sciences, mathematics and statistics	field-05.pdf
Information and Communication Technologies (ICTs)	field-06.pdf
Engineering, manufacturing and construction	field-07.pdf
Agriculture, forestry, fisheries and veterinary	field-08.pdf
Health and welfare	field-09.pdf
Services	field-10.pdf

The first public version of this guide, written for advanced researchers and senior professionals, stays as it was at [version 1](#).

Part III: Setting up

16. Labs: containers now, launch scripts next

In plain words. How a container packs a whole laboratory into one box that runs the same on every computer, and what comes next.

What a container is

A container is an ordinary process given its own view of the file system, the process table and the network through Linux namespaces, and held to a share of processor and memory by control groups. It starts from an image built in layers, each layer a set of file changes identified by its hash. The same image gives the same environment on every laptop, which is exactly what a teaching or research lab needs: the lab stops depending on what happens to be installed on the machine in front of you.

Checking that your machine is ready

```
docker run --rm hello-world
docker compose version
```

For GPU work, install the NVIDIA Container Toolkit by NVIDIA's [installation guide](#), then:

```
sudo nvidia-ctk runtime configure --runtime=docker
sudo systemctl restart docker
sudo docker run --rm --runtime=nvidia --gpus all ubuntu nvidia-smi
```

The last command prints the same table as `nvidia-smi` on the host, from inside a container.

What exists, and what comes next

Three things run in containers or isolated environments today: project-ilm/linguistics-labs, a starter container for linguistics collaborators whose full specification the author is to provide; TransEg's compose stand-up, which brings up the local digital twin; and PANINIq, which needs only a Python virtual environment.

The next release of this package carries the lab launch scripts: one script per lab, the software dockerised and retrieved from the existing repositories rather than rebuilt. The labs follow the pattern of the reference lab: most of the research behind current AI, robotics and automation papers, at modest cost, aligned with courses at the level of GATE in computer science and in robotics and automation.

Part V: Working with the estate

26. GATE: sit it, whatever your age

In plain words. India's national engineering examination: what it covers, how to apply, and why it is worth sitting at any age.

The facts for 2027

The Graduate Aptitude Test in Engineering 2027 is organised by IIT Madras. There are 30 test papers, including Robotics and Automation (RA) for the first time; a candidate may sit one paper or an allowed pair of two. The regular registration, first due to close on 27 September 2026, has been extended to 5 October 2026 without a late fee, and the extended registration, with a late fee, closes on 12 October 2026, all on the portal gate2027.iitm.ac.in. Registration goes through DigiLocker for Indian nationals. The application correction window runs from 14 to 21 October 2026, the examination is held in February 2027, and results are due on 19 March 2027. There is no upper age limit: a candidate needs to be in the third or a later year of an undergraduate degree, or to have completed one. A GATE score is valid for three years and opens postgraduate admissions and recruitment in public-sector undertakings.

The official syllabi

- Computer Science and Information Technology (CS): [CS_GATE2027_Syllabus.pdf](#)
- Robotics and Automation (RA): [RA_GATE2027_Syllabus.pdf](#)
- Instrumentation Engineering (IN): [IN_GATE2027_Syllabus.pdf](#)
- General Aptitude, common to every paper (GA): [GA_GATE2027_Syllabus.pdf](#)

Where the CS syllabus meets this guide

CS syllabus section	Chapters
Engineering mathematics: discrete mathematics, linear algebra, probability	11 (codes and the Kraft inequality), 13 (lattices and fixed points), 15 (vectors, unitary matrices, eigenvalues)
Digital logic: Boolean algebra, circuits, number representation and computer arithmetic	9, 14
Computer organisation and architecture: instructions and addressing, data path and control unit, pipelining	13 (microprogrammed control), 14
Programming and data structures	the examples throughout, in Python and C
Algorithms: graph algorithms and complexity, P and NP	13 (graph colouring), 15 (Max-Cut)
Theory of computation: regular and context-free languages, automata, Turing machines	12 (rewrite systems), 13 (automata and grammars)
Compiler design: lexing, parsing, syntax-directed translation, intermediate code, local optimisation, dataflow analysis	13, directly

CS syllabus section	Chapters
Operating systems	8 (namespaces and control groups), 14 (the boot chain)
Computer networks	9 (network byte order); not otherwise covered
Databases	not covered

For Robotics and Automation, read the RA syllabus beside the estate repositories that meet it in practice: [zistgah/pench](#), the embodied cyler; [zistgah/transeg](#), the local digital twin; and [zistgah/jugaad28](#) and [zistgah/zamin](#) on the hardware side.

Courses and the reference lab

Courses at GATE level for both CS and RA, aligned with the reference lab and its dockerised launch scripts, are **planned** as the next step after this guide. They will be announced in the [zistgah](#) organisation.

27. Syllabus links, and our syllabus

In plain words. Links from your school or exam syllabus to the chapters that teach the same ideas, worked out on your own device, and a draft syllabus of our own.

A student's syllabus and this guide cover many of the same ideas under different names. This chapter links them without copying a word of any syllabus: the guide stores only the official address of each syllabus and its own list of keywords, and the matching happens on the reader's own device.

How the link works. Choose a board or exam; the page opens its official syllabus, from the publisher's own site. If the browser may read it directly, the page reads it; if the publisher does not allow that, which most do not, save the syllabus from the official page and drop the file on this page. Either way, the text is read in your browser, matched against the guide's own keywords, and every match becomes a link into the chapter that teaches it. Nothing is uploaded, stored or sent anywhere, and the syllabus remains the property of its publisher.

The boards and exams linked

Board or exam	Body	Levels	Region	Official page
GATE 2027, Computer Science and Information Technology	IIT Madras	Graduate	India	gate2027.iitm.ac.in
GATE 2027, Robotics and Automation	IIT Madras	Graduate	India	gate2027.iitm.ac.in
GATE 2027, Instrumentation Engineering	IIT Madras	Graduate	India	gate2027.iitm.ac.in
GATE 2027, General Aptitude	IIT Madras	Graduate	India	gate2027.iitm.ac.in
GATE 2027, every paper	IIT Madras	Graduate	India	gate2027.iitm.ac.in
CBSE, classes 9 to 12	Central Board of Secondary Education	Secondary, senior secondary	India	cbseacademic.nic.in
NCERT textbooks	National Council of Educational Research and Training	Classes 1 to 12	India	ncert.nic.in
ICSE, class 10	Council for the Indian School Certificate Examinations	Secondary	India	cisce.org

Board or exam	Body	Levels	Region	Official page
ISC, class 12	Council for the Indian School Certificate Examinations	Senior secondary	India	cisce.org
NIOS	National Institute of Open Schooling	Secondary, senior secondary	India	www.nios.ac.in
Cambridge IGCSE	Cambridge International Education	Secondary	International	www.cambridgeinternational.org
Cambridge International AS and A Level	Cambridge International Education	Senior secondary	International	www.cambridgeinternational.org
IB Diploma Programme	International Baccalaureate	Senior secondary	International	www.ibo.org
IB Middle Years Programme	International Baccalaureate	Middle, secondary	International	www.ibo.org
Advanced Placement courses	College Board	Senior secondary	United States	apcentral.collegeboard.org
Common Core State Standards	National Governors Association and CCSSO	Kindergarten to grade 12	United States	www.thecorestandards.org
AQA GCSE and A level	AQA	Secondary, senior secondary	United Kingdom	www.aqa.org.uk
Pearson Edexcel qualifications	Pearson	Secondary, senior secondary	United Kingdom	qualifications.pearson.com
OCR qualifications	OCR	Secondary, senior secondary	United Kingdom	www.ocr.org.uk
Australian Curriculum, version 9	ACARA	Foundation to year 10	Australia	v9.australiancurriculum.edu.au
Singapore national examinations	SEAB	Secondary, pre-university	Singapore	www.seab.gov.sg
Telangana Board of Secondary Education	Government of Telangana	Secondary	India, state board	bse.telangana.gov.in
Andhra Pradesh Board of Secondary Education	Government of Andhra Pradesh	Secondary	India, state board	bse.ap.gov.in
Maharashtra State Board	Government of Maharashtra	Secondary, higher secondary	India, state board	mahahsscboard.in

Board or exam	Body	Levels	Region	Official page
Karnataka School Examination and Assessment Board	Government of Karnataka	Secondary	India, state board	kseab.karnataka.gov.in
Tamil Nadu Directorate of Government Examinations	Government of Tamil Nadu	Secondary, higher secondary	India, state board	dge.tn.gov.in
Kerala SCERT	Government of Kerala	School	India, state board	scert.kerala.gov.in
West Bengal Board of Secondary Education	Government of West Bengal	Secondary	India, state board	wbse.wb.gov.in
Uttar Pradesh Madhyamik Shiksha Parishad	Government of Uttar Pradesh	Secondary, intermediate	India, state board	upmsp.edu.in
Board of Secondary Education, Rajasthan	Government of Rajasthan	Secondary, senior secondary	India, state board	rajeduboard.rajasthan.gov.in
Gujarat Secondary and Higher Secondary Education Board	Government of Gujarat	Secondary, higher secondary	India, state board	www.gseb.org
Punjab School Education Board	Government of Punjab	Secondary, senior secondary	India, state board	www.pseb.ac.in

The GATE papers' addresses were retrieved for this guide on 29 September 2026; the other addresses are the publishers' own sites, where the current syllabus is published. Each carries its review mark in the register:

Our syllabus, a draft for review

A standard syllabus built from this guide, level by level, with its own outcomes, chapters and laboratory work. It is a draft for the author's review, and each module carries its review mark in the register at the end of the guide.

Foundation, classes 6 to 8

Module	Title	Hours	Outcomes	In this guide	Laboratory
F1	Numbers inside the machine	6	Count in binary and convert to decimal; Explain why a computer needs only two symbols	Chapter 17	

Module	Title	Hours	Outcomes	In this guide	Laboratory
F2	Letters, scripts and your language	6	Show that every letter is stored as a number; Write a word in two scripts and say what stays the same	Chapter 18, Chapter 19	
F3	The sky and the calendar	4	Read sunrise, moonrise and the tithi for a day; Explain why calendars differ	Chapter 10	chakra, jyotish
F4	Making with a cycler, safely	4	Make a short story or picture with a cycler; Say what you checked before keeping the result	Chapter 24	genie

Secondary, classes 9 and 10

Module	Title	Hours	Outcomes	In this guide	Laboratory
S1	Binary, logic and circuits	10	Build truth tables for basic gates; Relate a gate to transistors	Chapter 17, Chapter 22	
S2	Encoding and reversible transliteration	8	Explain UTF-8; Run a Romenagri round trip and judge whether it is exact	Chapter 18, Chapter 19	hindawi
S3	Programming in your own language	10	Compile and run a Hindi program; Find the same names in the debugger	Chapter 21	hindawi, urdu
S4	Using AI and checking it	6	Use any AI on a task and record each step; Tell an AI's claim from a verified result	Chapter 24, Chapter 8	alam

Senior secondary, classes 11 and 12

Module	Title	Hours	Outcomes	In this guide	Laboratory
H1	Grammar as a formal system	10	Write a regular expression and a small grammar; Read a Pāṇinian rule as a rewrite rule	Chapter 20, Chapter 21	tajziya
H2	From source code to machine	12	Trace a program from source through compiler, assembler and linker; Read symbols in an ELF file	Chapter 21, Chapter 22	hindawi
H3	Open source, provenance and contributing	6	Make a commit, open a pull request and pass a gate; Explain a hash, a manifest and a DOI	Chapter 25	rahnuma
H4	A project seeded from FAKIR	15	Choose a domain, a language and a layer and state a gate; Deliver the work with its evidence	Chapter 29	fakir

Undergraduate and GATE

Module	Title	Hours	Outcomes	In this guide	Laboratory
U1	Theory of computation and compilers	30	Automata, grammars and parsing; Build and debug a compiler pipeline	Chapter 20, Chapter 21, Chapter 26	hindawi , panini_by_claude
U2	Digital logic and architecture	30	From gates to a processor; Ternary logic and its trade-offs	Chapter 22, Chapter 26	pratik , zamin
U3	Quantum and physical computing	20	Qubits, gates and QAOA; Oscillator networks and Ising problems	Chapter 23	paniniq

Postgraduate and research

Module	Title	Hours	Outcomes	In this guide	Laboratory
P1	Research on the open questions	60	Choose an open question from Appendix C or D; Publish a measured, reproducible result	Appendix C, Appendix D, Appendix F	paniniq, tajziya

Professional

Module	Title	Hours	Outcomes	In this guide	Laboratory
Q1	Adopting a component	12	Run a component's gate; Integrate it with its contract and provenance intact	Chapter 15, Chapter 25, Appendix E	dhancha, humanesque

Appendices

E. For practitioners: teachers, policy makers, communities, makers and engineers

In plain words. A guide for teachers, policy makers, community workers, artists and engineers: where to start and how to take part.

Teachers and lab builders. Start from the reference lab, the GATE chapter and the component cards, which double as laboratory sheets: each says what to install, what to type and what a student should see (Chapters 16, 26 and 15). The reader views let each student follow a path suited to their age and background, and remember what they have read.

Policy makers and administrators. The questions that matter are in Chapter 10 and Appendix C: sovereignty in its testable technical sense, the ability to rebuild, verify, modify and run (Chapter 21); provenance that makes claims checkable; and infrastructure that leaves capability with communities. The executable contracts of the repositories show what verifiable governance of software looks like in practice.

Community workers and health practitioners. Project VIKRAM's sections cover education, health, livelihood, justice, environment and habitat, food and agriculture, and peace; its health section accepts submissions as issues. The Urdu edition and the Brahmi hub show work in people's own scripts; CHAKRA and Jyotish run offline.

Artists, writers and makers. The cyclers produce print, film, sound, immersive work, embodied work and the record; matba turns posters into a sealed book; GENIE writes and paints; PANINIphy turns a physical idea into parts. Every cycle is recorded as method, so a piece of work can be repeated, shared and credited.

Engineers in industry. Adopt one component at a time: read its card, clone it, run it, read its `CONTRACT.md` and its gate, check its DOI and provenance, then integrate it through its interface. † The notes give the full adoption ladder: the canonical artifact, a one-page explanation, the nearest industry analogue and the exact difference, a working demonstration, the repository, the DOI and timestamp, the interface, a case study, and the commercial route.

Contributors. Fork, open an issue, send a pull request. Every repository's gate runs on your change; every claim must point to its evidence; the AI you use is welcome, and its report that something passed is not the evidence that it did.

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